SoundnessBench: Can Your AI Scientist Really Tell Good Research from Bad?

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Abstract

Autonomous AI research agents aim to accelerate scientific discovery by automating the research pipeline, from hypothesis generation to peer review. However, existing benchmarks fail to address a fundamental bottleneck: the ability of Large Language Models to judge the viability of a research idea **before** expending time and computational resources. While LLMs excel at generative tasks like coding and writing, critical soundness judgment is a distinct cognitive challenge. Without a reliable "first-gate" filter, autonomous agents risk scaling flawed methodology rather than accelerating meaningful science. In this work, we introduce **SoundnessBench**, a curated benchmark of 1,099 real research proposals, grounded in and verified against published source papers. Our evaluation of 12 frontier LLMs reveals a pervasive **optimism bias**: models consistently rate proposals as sound regardless of their underlying quality. We find that aggressive prompting does not improve discrimination; it largely shifts errors from false positives to false negatives. Our results point to a deeper capability limitation in judging scientific soundness, indicating that current LLMs are not yet reliable as standalone gatekeepers for scientific rigor.

1 Introduction

Autonomous AI research agents are rapidly advancing in technical sophistication. Recent frameworks can autonomously generate hypotheses, implement code, execute experiments, and draft full manuscripts with minimal human oversight (Lu et al., 2024; Yamada et al., 2025; Schmidgall et al., 2025; Karpathy, 2026). While this progress suggests a future of accelerated discovery, it also exposes a critical, under-addressed bottleneck: *scientific triage*. Before a single line of code is written, a researcher must determine if a proposed idea and its experimental design are methodologically sound. In human scholarship, this "first-gate" judgment is the primary defense against wasting months of labor and substantial computational resources.

The Soundness Gap. Existing evaluations of AI agents largely overlook this decision point. Current benchmarks (Chan et al., 2024; Starace et al., 2025; Wu et al., 2025; Lupidi et al., 2026) primarily emphasize *execution* to reproduce existing results or deliver runnable artifacts. They do not test whether an agent can critically evaluate the viability of a proposal in the first place. This leaves a fundamental question unanswered:

> Can LLMs reliably reject flawed research designs before they are executed?

Without a robust upfront filter, autonomous agents do not necessarily accelerate science; they risk scaling "bad science" by automating the pursuit of unsound hypotheses.

We define **scientific soundness** not as the eventual "impact" or "novelty" of a paper, but as its *methodological integrity*: given the hypothesis, is the experimental design capable of testing it rigorously? In the lifecycle of an autonomous agent, soundness judgment serves as the *pre-computation check*. Without this capability, agents risk entering a "hallucination-implementation loop," where they generate logically flawed experiments that appear structurally correct but are scientifically dead on arrival. By isolating soundness from

Table 1: **Comparison with related research-agent benchmarks. Stage** indicates when scientific judgment is made (pre-execution, execution, or post hoc). **Benchmark Task** indicates what is judged, and **Evaluation Input** indicates evidences provided to the evaluator. Most prior benchmarks evaluate execution outcomes or post-hoc signals rather than methodological validity. SoundnessBench is the only benchmark here that combines **pre-execution evaluation**, **direct methodological-soundness judgment**, and **proposal-only input**.

Benchmark	Stage	Benchmark Task	Evaluation Input	Ground Truth Source	Soundness Focus	Pre-Exec. Judgment
MLE-Bench	Exe.	Engineering	Task desc. + Exp. Results	Kaggle outcome	✗	✗
PaperBench	Exe.	Replication	Full paper + Exp. Results	Human rubric	✗	✗
InnovatorBench	Exe.	Research loop	Task desc. + Exp. Results	Outcome metric	✗	✗
AIRS-Bench	Exe.	Multi-task	Task desc. + Exp. Results	Expert annotation	✗	✗
ResearchGym	Exe.	Research loop	Task desc. + Exp. Results	Outcome metric	✗	✗
Si et al. (Si et al., 2025b)	Pre-exec.	Novelty judge	Proposal	Human study	✗	✓
Hindsight	Post-hoc	Impact pred.	Idea	Future citation/venue	✗	✗
RINoBench	Pre-exec.	Novelty judge	Idea + related works	Expert label	✗	✓
Tong et al. (Tong et al., 2026)	Post-hoc	Impact pred.	Title + abstract	Future citation/venue	✗	✗
SoundnessBench	**Pre-exec.**	**Methodological soundness**	**Proposal**	**Expert label**	✓	✓

downstream results, we test the model's ability to identify fatal flaws (such as improper baselines, data leakage, or mismatched metrics) before they incur high execution costs.

However, the utility of such a check is predicated on its stability. This triage is further complicated by known model behaviors. Prior work on sycophancy and prompt fragility suggests that LLM judgments are highly sensitive to how a question is framed (Sharma et al., 2023; Elkins & Chun, 2025; Geng et al., 2026). If scientific evaluation inherits these vulnerabilities, autonomous agents may suffer from inconsistent filtering, alternating between reckless optimism and destructive skepticism based on minor prompt variations.

In this paper, we introduce **SoundnessBench**, a large-scale benchmark for pre-execution scientific judgment. While existing datasets often rely on small-scale human annotations or synthetic tasks, SoundnessBench is reconstructed from the ICLR public history. We processed over 35,209 initial submissions and 137,940 expert reviews to identify a high-signal subset of 1,099 hypothesis–experiment pairings. This scale covers 16 distinct ML subfields ranging from Reinforcement Learning to Generative Modeling to ensure that the benchmark evaluates "generalist" scientific reasoning rather than niche pattern matching. Our pipeline utilizes reviewer-agreement filtering, near-verbatim proposal extraction, and a retrieval-backed atomic-claim auditing process to ensure that the extracted proposals remain faithful to the original manuscripts.

Our evaluation of 12 frontier LLMs reveals a pervasive **optimism bias**. Under standard prompting, models frequently classify low-soundness proposals as high-soundness, yielding a mean false-positive rate of 74.0%. Conversely, when subjected to aggressive prompting designed to encourage skepticism, the models over-correct: while the false-positive rate drops to 19.9%, high-soundness recall collapses to 36.1%. These results suggest that current frontier models are not yet reliable as standalone scientific critics; their judgments are both inaccurate on flawed methodology and highly unstable under different prompt framings.

Our contributions are:

- **SoundnessBench:** A diverse benchmark of 1,099 validated research proposals. This represents one of the largest curated datasets of its kind, spanning 8 years of top-tier ML submissions and 16 sub-disciplines, providing a statistically significant measure of model "common sense" in research.
- **A High-Precision Curation Pipeline:** We detail a rigorous multi-stage pipeline that moves beyond simple scraping. Our process includes: (1) *Expert-Agreement Filtering* (utilizing 4,391 human-written reviews), (2) *Atomic-Claim Auditing* to prevent "benchmark leakage," and (3) *Outcome Masking* to ensure models judge the methodology rather than predicting the paper's eventual acceptance.

- **Quantifying the "Optimism-Fragility" Tradeoff:** We provide an empirical study of 12 frontier LLMs, documenting a systemic failure to identify methodological flaws. We characterize the "optimism bias" and demonstrate that current "AI Scientists" are highly sensitive to prompt framing, making them brittle as autonomous decision-makers.

2 SoundnessBench: Benchmark Reconstruction

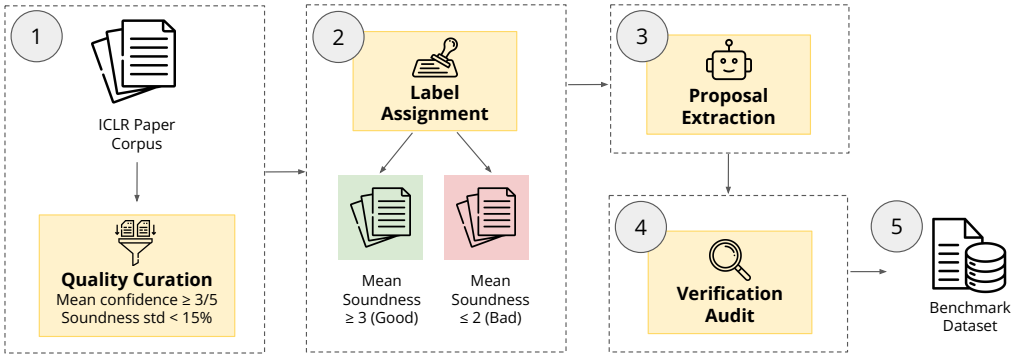

Figure 1: **SoundnessBench pipeline:** (1) collect ICLR papers with reviewer metadata and filter for high reviewer agreement; (2) derive high/low-soundness labels; (3) extract near-verbatim research proposal without revealing experimental results; (4) audit extraction fidelity with retrieve-then-verify atomic claims; and (5) assemble the final benchmark.

Overview and Motivation. In human research, the first gate happens before implementation: researchers and advisors ask whether a hypothesis is well-posed and whether the planned experiment can truly validate or falsify the claim. This early judgment decides which ideas deserve months of engineering effort and compute.

SoundnessBench targets this first-gate setting for LLM agents. Most existing LLM benchmarks evaluate downstream skills (writing, coding, or reviewing completed papers), with limited direct testing of whether an agent can reject weak designs before execution. SoundnessBench addresses this gap by isolating pre-execution rigor assessment, grounding labels in peer-review outcomes, and auditing extraction faithfulness so each instance is traceable to source evidence. In short, it probes how well current LLM-based agents can make early-stage soundness judgments from proposal text alone.

Challenges and Design Choices. Constructing a proposal-only soundness benchmark from real submissions required us to address three challenges up front. First, *label reliability*: reviewer judgments can be noisy and mixed, so we use ICLR reviewer soundness scores as a proxy, keep only high-agreement reviews, and remove ambiguous middle-score cases. Second, *information leakage*: completed papers contain results that can shortcut true first-gate assessment, so we exclude experimental outcomes and keep only proposal components. Third, *extraction fidelity*: proposal extraction can introduce drift, so we preserve near-verbatim wording and add a retrieval-backed atomic-claim verification audit.

We chose these design decisions to maximize reliability, task faithfulness, and traceability at the same time. Compared with prior benchmarks that emphasize downstream tasks or rely primarily on post-hoc annotation, SoundnessBench explicitly combines peer-review-grounded labels, proposal-only inputs, and evidence-audited extraction, which enables a cleaner test of pre-execution scientific judgment.

Benchmark Reconstruction. We build SoundnessBench in five steps, shown in Fig. 1.

- **Step 1: Data Collection.** We collect papers from the ICLR corpus, covering major ML areas such as RL, generative modeling, NLP, optimization, and computer vision (16 sub-

domains in total). We start from 2022 because earlier releases do not consistently provide reviewer soundness (or closely related) scores needed for filtering and label derivation. This multi-year, multi-subfield pool helps reduce narrow-domain bias. From the collected papers, we remove desk-rejected papers, since desk rejections may reflect factors other than scientific soundness. Then, to reduce label noise, we retain only papers with strong reviewer agreement: mean reviewer confidence of at least 3 and soundness-score standard deviation below 15%.

- **Step 2: Label Assignment.** We assign labels from the mean soundness score: pairs with scores ≥ 3 are labeled *high* soundness, whereas pairs with scores ≤ 2 are labeled *low* soundness. This removes ambiguous middle-score cases and improves class separation.
- **Step 3: Proposal Extraction.** From filtered papers in Step 2, we extract a research proposal including the abstract, related work, risk factors, hypothesis, and experiment design directly from source PDFs, while excluding experimental results. The proposal format follows The AI Scientist-v2 (Yamada et al., 2025). We preserve near-verbatim wording to reduce paraphrase drift and keep inputs aligned with the original submissions. We use a strong long-context LLM (Gemini 2.5 Pro) with a task-specific prompt (Supplementary Sec. A.1). Example extractions are provided in Supplementary Sec. A.4.
- **Step 4: Verification Audit.** To ensure the extracted proposals faithfully match source PDFs and to minimize extraction errors from Step 3, we add a verification audit. The detailed process is presented in Alg. 1. We check extraction faithfulness with a retrieval-backed atomic-claim score. Following prior work on atomic factuality and evidence-grounded verification (Min et al., 2023; Thorne et al., 2018), we decompose each extracted hypothesis–experiment pair into atomic claims, retrieve supporting passages, and verify each claim against the source paper (Supplementary Sec. A.2, Sec. A.3, and Sec. A.5). We set $\tau = 0.7$, chunk size $s = 3000$, chunk overlap $o = 200$, and top-k $= 3$. In our current pool, 66.93% of candidates pass this filter, yielding a benchmark with auditable hypothesis–experiment designs. The mean support ratio is shown in Fig. 2.
- **Step 5: Final Benchmark Dataset.** The final benchmark contains 1,099 research proposals, including 458 low-soundness and 641 high-soundness instances. Detailed benchmark statistics are presented in Fig. 2.

Algorithm 1 Verification Audit

Require: Source PDF $\mathcal{P}$, extracted pair $\hat{p} = (h, e)$, threshold τ, chunk size s, chunk overlap o, k
Ensure: Fidelity score $s \in [0, 1]$, and binary pass/fail decision

```
T ← ExtractText(P)
C ← Chunk(T, s, o)
A ← DecomposeAtomicClaims(p̂)
supported ← 0
for each claim c ∈ A do
    E ← BM25(c, C, top-k = k)
    v ← LLMVerify(c, E, evidence_only = True)
    if v = YES then
        supported ← supported + 1
    end if
end for
s ← supported / |A|
return s ≥ τ
```

3 Evaluation

This section is organized as follows: we first describe the evaluation protocol, models, and metrics (Sec. 3.1); then present the main results under standard prompting (Sec. 3.2); and finally analyze how aggressive prompting changes model behavior and calibration (Sec. 3.3).

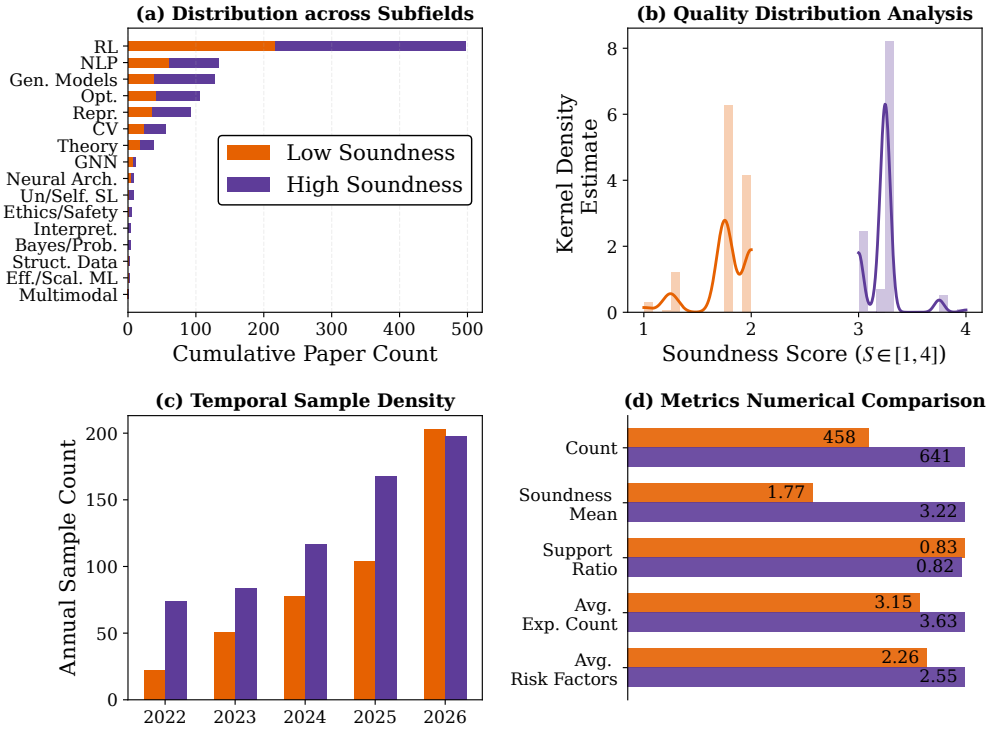

Figure 2: **SoundnessBench dataset statistics.** The benchmark includes 1,099 proposals, including 458 low-soundness and 641 high-soundness instances. **(a)** Subfield distribution across papers reflects ICLR corpus composition. **(b)** Soundness score density shows separation between low-soundness ($S < 2$, mean $= 1.77$) and high-soundness ($S > 3$, mean $= 3.22$) groups, supporting the chosen label boundary. **(c)** Temporal coverage spans ICLR 2022–2026. **(d)** Low- and high-soundness pair-count statistics in SoundnessBench.

3.1 Evaluation Setup

We evaluate frontier large language models on SoundnessBench, which is built from real ML research submissions. In each test, we provide a research proposal to an LLM with a carefully designed system prompt. The model is asked to classify the proposal as high or low soundness and provide an explanation.

Models. We evaluate a diverse set of **12 frontier models** spanning small and large scales, non-reasoning and reasoning, and both closed- and open-source models. The evaluated set includes GPT models (GPT-4o, GPT-5.4-Mini, GPT-5.4), Claude models (Claude-Opus-4.6, Claude-Sonnet-4.6), Gemini models (Gemini-2.5-Pro, Gemini-3-Flash, Gemini-3.1-Pro), Qwen models (Qwen-3.5-27B, Qwen-3.5-122B-A10B), a LLaMA model (LLaMA-3.3-70B), and a Kimi model (Kimi-Linear-48B-A3B).

Inference Setup. For closed-source models, we run inference through the corresponding provider APIs. For open-source models, we deploy model servers with vLLM on 2×NVIDIA H200 GPUs. Unless otherwise noted, we use a consistent decoding configuration across models with `max_tokens=8192` and `temperature=0.2`; other generation parameters follow framework/provider defaults.

Prompting Protocol. We evaluate each model under two prompting regimes:

- **Standard prompt.** This is a standard evaluation mode and models are asked to assess the soundness of a research proposal without an explicit prior bias.

Table 2: Summary evaluation metrics across 12 models under standard and aggressive prompting. Low R and High R denote recall on low- and high-soundness proposals. Macro F1 is the unweighted average of per-class F1 scores. Best Macro F1 per condition is **bolded**. † denotes reasoning model.

	Standard Prompt			Aggressive Prompt		
Model	**Low R ↑**	**High R ↑**	**Macro F1 ↑**	**Low R ↑**	**High R ↑**	**Macro F1 ↑**
Open-source Models						
LLaMA-3.3-70B	2.0	99.4	38.9	39.3	76.1	57.5
Kimi-Linear-48B-A3B	13.8	84.6	44.6	50.2	45.8	47.5
Qwen3.5-27B†	23.6	92.5	55.1	83.8	32.4	52.6
Qwen3.5-122B-A10B†	26.6	90.6	56.3	95.6	16.8	44.7
Closed-source Models						
GPT-4o†	5.5	98.9	42.3	87.1	25.0	48.5
GPT-5.4-mini	50.7	76.6	63.8	100.0	0.2	29.7
GPT-5.4†	64.6	74.6	**69.7**	100.0	0.0	29.5
Gemini-2.5-Pro†	13.8	98.4	49.8	73.4	55.1	62.7
Gemini-3-Flash	19.7	98.1	54.5	76.4	60.4	67.1
Gemini-3.1-Pro†	26.0	96.1	58.4	89.1	35.9	57.0
Claude-Sonnet-4.6†	37.2	94.8	65.3	94.4	18.8	46.0
Claude-Opus-4.6†	28.2	97.2	60.5	72.2	66.4	**68.6**
Mean	26.0	91.8	54.9	80.1	36.1	49.3

- **Aggressive prompt.** Motivated by the optimism bias observed under the standard prompt, we instruct models to default to *low soundness* unless the idea and experimental design are clearly strong and well-justified.

This setup allows us to probe whether models possess a stable notion of scientific soundness, or whether their judgments are sensitive to prompt framing. Data-construction prompts are documented in Supplementary Sec. A, and additional evaluation analyses are provided in Supplementary Sec. B.

Metrics. We report confusion matrices over the binary classification task, evaluating the ability to (i) reject low-soundness ideas and (ii) correctly identify high-soundness ones. An ideal model should concentrate on the diagonal entries (low→low and high→high), with both low false positives on low-soundness proposals and low false negatives on high-soundness proposals. Off-diagonal errors indicate failure modes: optimism bias (low→high) or over-conservatism (high→low). We also report class-wise recall for both labels (Low R and High R) and Macro F1 (the unweighted average of per-class F1 scores).

3.2 Main Results: A Broad Optimism Bias in Scientific Judgment

Across LLM families, we observe a pervasive optimism bias: models consistently rate proposals as sound regardless of their underlying quality. As summarized in Tab. 2 and visualized in Fig. 3, many proposals labeled as low soundness are predicted as high soundness. Specifically, under standard prompting, mean low-soundness recall is 26.0% (equivalently, a 74.0% false-positive rate), while mean high-soundness recall remains 91.8%. At the model level, 9/12 models (including Gemini-3.1-Pro and Claude-Opus-4.6) label more than 70% of low-soundness proposals as high soundness. The strongest optimism appears in LLaMA-3.3-70B and GPT-4o, which classify 98.0% and 94.5% of low-soundness proposals as high soundness. This pattern indicates that model judgments are often skewed toward approval even when proposal quality is limited. In practice, many default model configurations behave as permissive first-gate reviewers. Instead of consistently filtering weak proposals, they approve a broad set of candidates, which can lead to avoidable compute spend and weaker downstream experimentation.

A few frontier LLMs reduce optimism bias, but at the cost of missing more high-soundness proposals. For example, GPT-5.4 and GPT-5.4-mini are more conservative, but their high-soundness recall drops to 74.6% and 76.6%, respectively. This suggests a calibration trade-off in current

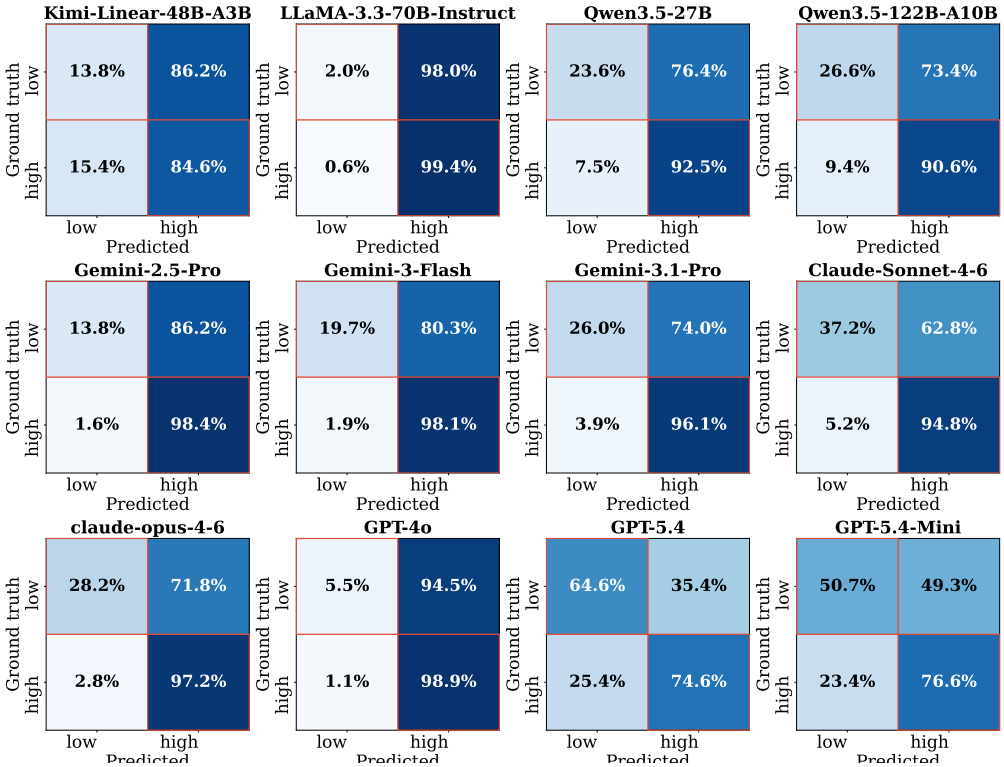

Figure 3: Confusion matrices under the standard prompt across 12 evaluated models. Main message: many models are over-optimistic by default. The mean false-positive rate on low-soundness proposals is 74.0% (9/12 models exceed 70%). This pattern appears across model families in this evaluation setting.

frontier LLMs for scientific judgment: stricter filtering can suppress false approvals, but it also increases the risk of rejecting promising ideas.

This optimism bias remains robust across the tested interventions. As shown in Supplementary Sec. B.1, increasing model scale within a single family does not eliminate this pattern. Supplementary Sec. B.2 further indicates that instruction tuning by itself is also insufficient in this setting. Taken together, these results suggest that the limitation is not restricted to one model size or one tuning stage, and that more targeted calibration strategies may be needed for reliable scientific judgment.

3.3 Under Aggressive Prompting: Bias Shifts Toward Over-Conservatism

Aggressive prompting does not improve discrimination; it mainly shifts errors from false positives to false negatives. We observe widespread optimism bias in Sec. 3.2. Motivated by this, we test whether aggressive prompting can improve LLM scientific assessment. To probe whether prompting alone can correct this bias, we keep the same models and task, but make the instruction stricter: default to *low soundness* unless the idea and experimental design are clearly strong and well-justified. Additional analyses for this setting are presented in Supplementary Sec. B.1 and Sec. B.2. As shown in Tab. 2 and Fig. 4, models generally become more conservative: low-soundness rejection improves, but high-soundness identification drops. The trade-off is quantitatively large. Mean low-soundness false positives drop from 74.0% (standard prompt) to 19.9% (aggressive prompt), and 10/12 models are below 30%; meanwhile, mean high-soundness recall drops from 91.8% to 36.1%, with 7/12 models below 40%. At the extreme, GPT-5.4 and GPT-5.4-Mini move to near-always-low behavior under aggressive prompting: low-soundness false positives are 0% for both, while high-soundness recall falls to 0.0% and 0.2%, respectively. Other models show the same pattern

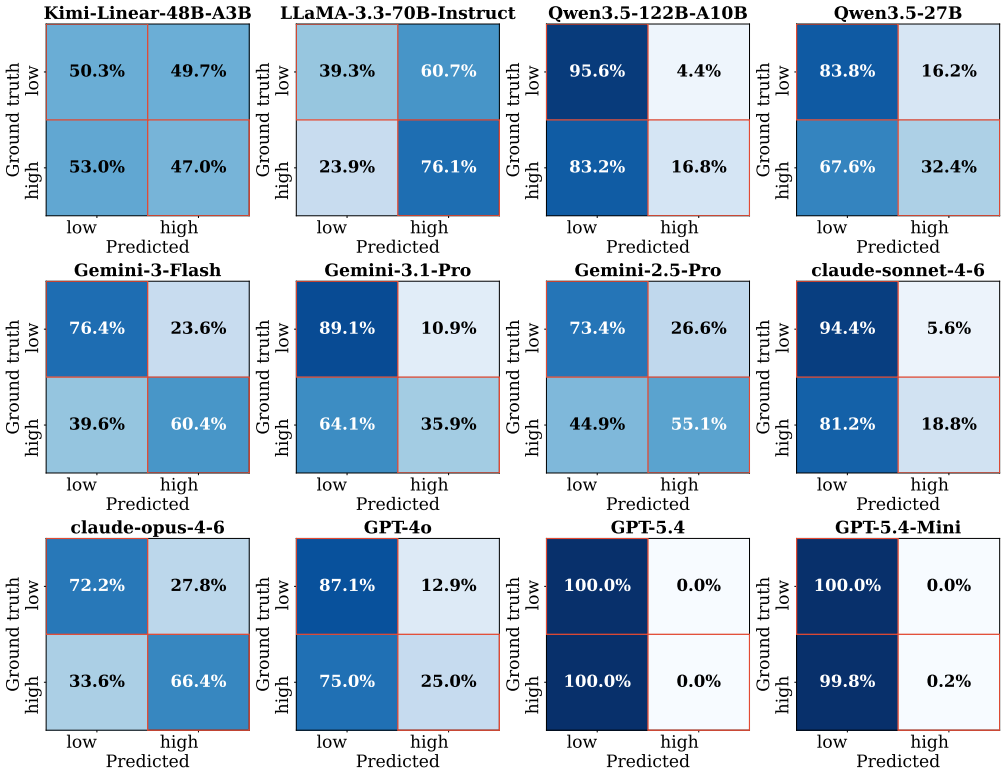

Figure 4: Confusion matrices under the aggressive prompt across 12 evaluated models. Main message: optimism bias often shifts toward over-conservatism. The mean false-positive rate on low-soundness proposals drops to 19.9% (10/12 models are below 30%), but recall on high-soundness proposals also drops to 36.1% (7/12 models are below 40%). This illustrates strong prompt sensitivity in scientific-soundness judgment for the evaluated models.

in milder form; for example, Qwen3.5-122B-A10B and Claude-Sonnet-4.6 strongly reject low-soundness ideas (95.6% and 94.4% rejection), but correctly identify only 16.8% and 18.8% of high-soundness proposals. Consistent with this, mean Macro F1 decreases from 54.9 to 49.3.

Taken together, these results point to a deeper capability limitation: model judgments are unstable under prompt changes. Prompt sensitivity varies substantially across models. Remarkably, GPT-5.4 has the best Macro F1 under standard prompting (69.7%) but drops sharply under aggressive prompting (29.5%), whereas Claude-Opus-4.6 changes much less (60.5% vs. 68.6%). This cross-model variation, together with the robustness analyses (see Supplementary Sec. B.2 and Sec. B.1), suggests that the optimism bias is prompt-sensitive, persists across tested model scales, and is not removed by instruction tuning within the examined family.

Key Takeaway 1. Frontier LLMs show a broad optimism bias: models consistently rate proposals as sound regardless of their underlying quality.

Key Takeaway 2. Aggressive prompting does not improve discrimination; it mainly shifts errors from false positives to false negatives.

Key Takeaway 3. Taken together, results point to a deeper capability limitation in judging scientific soundness: current LLMs are not yet reliable as standalone gatekeepers for scientific rigor.

4 Related Work

Autonomous AI Research Agents. Recent AI scientist agents aim to automate large parts of the research pipeline. *The AI Scientist* (Lu et al., 2024) and *The AI Scientist-v2* (Yamada et al., 2025) are early systems that attempt this process by autonomously generating ideas, writing code, running experiments, and drafting papers. *Agent Laboratory* (Schmidgall et al., 2025) structures this process into literature review, experimentation, and report writing, with optional human feedback. Beyond papers, *autoresearch* (Karpathy, 2026) provides a simple open-source loop where an agent edits code, runs short real experiments, and keeps changes only when metrics improve. These systems show that LLM agents can execute most parts of research workflows. However, they do not directly test whether LLMs can judge if a research proposal (including the core idea, related work context, and experimental design) is worth pursuing *before* execution begins. Like human researchers triaging projects, this first-gate decision determines whether time and compute should be invested.

Evaluating Research Agents and Idea Quality. Recent studies suggest that LLM-generated ideas can be novel while being weaker on feasibility (Si et al., 2025b) and this gap can widen after implementation (Si et al., 2025a). At the system level, most current benchmarks focus on evaluating execution (Chan et al., 2024; Starace et al., 2025; Wu et al., 2025; Lupidi et al., 2026). Additionally, a few idea-evaluation benchmarks target dimensions such as impact and novelty (Jiang, 2026; Schopf & Färber, 2026), or model scientific taste from community-level signals (publication venue and citation outcomes) (Tong et al., 2026). These directions are important, but their targets differ from ours. In contrast, SoundnessBench targets a narrower upstream question: whether a research proposal is methodologically sound before results exist and whether it is a good test of the claimed idea.

LLMs as Peer Reviewers. Existing work studies LLMs as reviewers of completed papers. LLM assistance can improve review clarity and usefulness in human review workflows (Thakkar et al., 2025). Recent benchmarks also study the behavior and quality of LLM-based reviewing systems at scale (Huang et al., 2026; Yu et al., 2025). They study how models produce critiques and how closely their judgments track human assessments. However, this literature evaluates *post hoc* review, where the paper, results, and framing are already available. SoundnessBench instead studies *pre-execution* judgment from research proposal alone.

LLM Sycophancy and Fragile Judgment. Our work is also connected to studies of LLM sycophancy and fragile judgment. Previous work demonstrates that LLMs often align with user framing rather than ground truth, which can make them validate plausible but weak claims (Sharma et al., 2023). Other work shows that attempts to suppress this behavior can produce unstable skepticism or excessive refusal (Geng et al., 2026). Prompt-framing studies further show that small syntactic changes can flip model decisions even when content is unchanged (Elkins & Chun, 2025). These findings suggest that scientific criticism is not only a knowledge problem, but also a robustness problem. SoundnessBench tests both.

5 Conclusion

We introduced SoundnessBench to evaluate a core capability for autonomous science: deciding whether a hypothesis and experimental design are scientifically sound before running expensive experiments. Across the models we tested, judgments are often sensitive to prompt framing: under standard prompting many models over-approve weak proposals, while stricter prompting can push some models toward broad over-rejection. At this early gate in the research pipeline, such instability can misallocate compute and attention. We view SoundnessBench as a step toward safer and more useful AI research agents, and as a tool for diagnosing where model calibration breaks down. The observed optimism tendency and prompt sensitivity suggest that reliable scientific judgment will likely require targeted training or calibration approaches beyond prompting alone.

Limitations. Our ground truth relies on reviewer soundness scores, and the benchmark covers a bounded slice of ML papers. Expanding to broader venues, modalities (e.g., code and logs), and longitudinal settings is an important next step.

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Supplementary

We provide additional details about data-construction prompts and verification procedures, and present extended evaluation results that complement the main findings and support their robustness. These materials could not be included in the main paper due to space constraints. Additionally, **benchmark is provided in the submitted supplementary file.**

A Supplementary Material for Data Construction

A.1 Prompt to Extract Proposal

The prompt is designed to extract proposal-level content from each paper while explicitly excluding results and conclusions. The proposal format follows Yamada et al. (2025).

System Prompt to Extract Proposal

You are an expert at extracting structured research information from machine learning papers. Extract the research proposal in *AI Scientist v2* format, capturing the intended hypothesis and experimental design **before any experimental results**. The goal is to represent the research plan and claims prior to execution, not confirmed findings.
Output: Return **valid JSON only**. Do not include markdown or explanations.

Global Principles

- Extract only proposal-level content; exclude results and conclusions.
- Prefer verbatim spans from the paper when possible.
- Do not hallucinate or introduce structure not present in the paper.
- If uncertain, omit the information.
- Keep claims atomic and directly grounded in the source text.

Hypothesis

- Extract from Introduction or Method sections.
- Describe what the paper proposes or aims to test (1–2 sentences).
- Prefer verbatim phrasing.

Forbidden: "we show", "we demonstrate", "we find", "results show", "outperforms"
Allowed: "we propose", "we hypothesize", "we argue", "our approach aims to"

Experiments

- Describe experimental plan only (no results or numbers).
- Each entry must correspond to a distinct experiment group.

Description: goal of the experiment.
Method: datasets, models, baselines, setup (preserve exact names).
Metrics: list evaluation metrics (no values).

Related Work

- Summarize prior work and research gap.
- Do not include improvement claims.

Risk Factors

- Extract explicit limitations from the paper.
- Do not infer or invent new risks.

Abstract

- Keep only problem, motivation, and proposed method.
- Remove results, numbers, and comparisons.

User Prompt to Extract Proposal

Input

- **Abstract:**

  `{abstract}`

- The attached file is the **full paper PDF**.

Use both the abstract and the PDF to extract the research idea.

Instructions

- Extract the **Short Hypothesis** from the **Introduction or Method sections**, not from results or conclusions.
- All **Experiment fields** must reflect the **experimental plan**, not observed outcomes.
- Preserve **dataset names, tasks, and baselines** when describing experiments.
- Create **multiple experiment entries** if the paper evaluates multiple tasks or benchmarks.

Output Format (JSON)

```
{
  "Name": "<short slug identifying the research contribution>",
  "Title": "<exact paper title>",
  "Short Hypothesis": "<proposed claim or method | no outcomes>",
  "Related Work": "<prior work and research gap>",
  "Abstract": "<problem + motivation + proposed method only>",
  "Experiments": [
    {
      "Description": "<goal of this experiment group>",
      "Method": "<datasets/tasks, model setup, baselines>",
      "Evaluation Metrics": "<metrics used>"
    }
  ],
  "Risk Factors and Limitations": [
    "<explicit limitation sentence from the paper>"
  ]
}
```

Critical Constraints

- Do **not** include results or conclusions.
- Do **not** include numerical values or measured performance.
- Do **not** reveal whether the hypothesis was confirmed.
- Do **not** generalize dataset names.
- Do **not** merge distinct experiments into a single entry.

A.2 Prompt to Decompose Atomic Claims from Proposal

This prompt decomposes each proposal into minimal, independently verifiable factual claims so that each statement can be assessed clearly and grounded in source text. The detailed process is presented in Alg. 1. We compute extraction faithfulness with a retrieval-backed atomic-claim score by decomposing each hypothesis–experiment pair into atomic claims, retrieving supporting passages, and verifying each claim against the source paper, following prior work on atomic factuality and evidence-grounded verification (Min et al., 2023; Thorne et al., 2018). We use GPT-5.4 for this process.

System Prompt to Decompose Atomic Claims from Proposal

You are a precise scientific analyst. Your task is to decompose a research hypothesis and experimental plan into atomic, independently verifiable claims.

User Prompt to Decompose Atomic Claims from Proposal

Task. Decompose the following extraction into **atomic factual claims** that are directly verifiable from the paper text.

Rules

- Output **one complete claim per line**.
- Do not include numbering, bullets, prefixes, section headers, or commentary.
- Do not output incomplete fragments.
- Do not introduce synthetic labels (e.g., "Experiment 1", "Experiment 2").
- Keep claims literal and close to wording likely present in the paper.

Scope of Claims Include only claims about:

- proposed method or hypothesis
- planned evaluation setup
- datasets, tasks, and baselines
- evaluation metrics

Exclude:

- any result or outcome claims (e.g., "improves", "outperforms", "achieves state-of-the-art")

Input Extraction

```
{extraction_proposal}
```

A.3 Prompt to Verify Atomic Claims

This prompt evaluates each claim–evidence pair and produces a concise YES/NO judgment so that verification decisions remain explicit, consistent, and grounded in the provided textual evidence. We use Gemini-2.5-Flash for this process.

System Prompt to Verify Atomic Claims

You are a rigorous scientific fact-checker. Your task is to verify whether a claim is **directly supported** by the provided evidence from a paper.

User Prompt for Atomic Claim Verification

Input Claim

```
{claim}
```

Evidence (excerpts from the paper)

```
{evidence}
```

Task
Determine whether the claim is **directly supported** by the provided evidence.

Output Format
Reply with exactly:

```
YES/NO: <one-sentence explanation>
```

A.4 Example of Extracted Proposals

We provide two extracted-proposal examples from SoundnessBench: one high-soundness case and one low-soundness case. The benchmark contains 458 low-soundness and 641 high-soundness instances in total. The format of these proposals follows (Yamada et al., 2025).

Example: High-Soundness Proposal

Paper id: FUaDMRVrbS
Identifiability for Gaussian Processes with Holomorphic Kernels
Year: 2025
Short Hypothesis
A novel theoretical framework, based on the property of kernels being holomorphic around zero, can determine the identifiability of parameters in a wide range of GP kernels (e.g., squared exponential, periodic, rational quadratic) and their complex combinations, filling a gap left by methods that only apply to Matérn-type kernels.
Related Work
[Content omitted for brevity]
Abstract
[Content omitted for brevity]
Experiments
Experiment 1
Description: To empirically support the theoretical results on parameter identifiability for several individual holomorphic kernels.
Method: The experiment involves estimating parameters for individual Squared Exponential (SE), Damped Periodic (DPer), Periodic (Per), Rational Quadratic (RQ), and Cosine kernels. This is done by fitting a GP model to synthetically generated data of increasing sample sizes ($n \in \{500, 1000, 2000, 5000\}$) with added Gaussian noise. The parameters are estimated using Maximum Likelihood Estimators (MLEs).
Evaluation Metrics: Convergence of the Maximum Likelihood Estimators (MLEs) for each parameter over 100 replicates.

Experiment 2
Description: To empirically support the theoretical results on parameter identifiability for a complex combined kernel, specifically the one from Equation (2) motivated by the Mauna Loa CO2 dataset.
Method: A GP with the combined kernel from Equation (2) is fitted to synthetically generated data designed to mimic the properties of the Mauna Loa dataset over a 45-year time interval. The experiment uses increasing sample sizes ($n \in \{50, 100, 200, 500\}$), and the Maximum Likelihood Estimators (MLEs) for the kernel's 10 parameters are computed.
Evaluation Metrics: Convergence of the Maximum Likelihood Estimators (MLEs) for each parameter in the combined kernel over 100 replicates.
Risk Factors and Limitations

- First, while establishing the identifiability of kernel parameters is a critical step, it does not necessarily guarantee the consistency of the MLE.
- Second, extending our theoretical framework to encompass non-stationary kernels ... remains a largely open problem.
- Third, another intriguing direction involves extending findings to infinitely differentiable kernels not holomorphic near 0.

Scores:

- Soundness: 3.25
- Support Ratio: 1.0
- Rigor Bucket: high

Example: Low-Soundness Proposal

Paper id: h9ThYkkgSD4
Activation Function: Absolute Function,One Function Behaves more Individualized
Year: 2023
Short Hypothesis
Using the absolute value function, y=—x—, as a neural network activation will result in more ïndividualizedřepresentations by preserving information from negative inputs, unlike ReLU. This property is hypothesized to make it more suitable for generation tasks and potentially less prone to overfitting compared to ReLU and Leaky ReLU.
Related Work
[Content omitted for brevity]
Abstract
[Content omitted for brevity]
Experiments
Experiment 1
Description: To evaluate the performance of the absolute activation function against standard activations in a fully-connected neural network on an image classification task.
Method: A 5-layer fully-connected neural network will be trained on the MNIST dataset. The performance of the absolute function will be compared against ReLU and Leaky ReLU as baselines. The experiment will use the Adam optimizer and SparseCategoricalCrossentropy loss. The effect of different batch sizes will also be investigated.
Evaluation Metrics: ['Training Accuracy', 'Validation Accuracy', 'Training Loss', 'Validation Loss']

Experiment 2
Description: To evaluate the performance of the absolute activation function in a convolutional neural network (CNN) on an image classification task and to visualize the resulting feature maps.
Method: A CNN with 3 convolutional layers and 2 fully-connected layers will be trained on the MNIST dataset. The absolute function will be compared against ReLU and Leaky ReLU baselines. The experiment will explore the effect of varying batch sizes (32, 64, 128). Additionally, the intermediate activations from the convolutional layers will be visualized for both the absolute function and ReLU to qualitatively compare the learned features.
Evaluation Metrics: ['Training Accuracy', 'Validation Accuracy', 'Training Loss', 'Validation Loss', 'Qualitative visualization of intermediate activations']

Experiment 3
Description: To test the hypothesis that the ïndividualizationp̈roperty of the absolute function makes it well-suited for image generation tasks using an autoencoder.
Method: A functionally separate autoencoder, with a convolutional encoder and a corresponding decoder, will be trained on the MNIST dataset. The experiment will compare the image generation quality when using the absolute function in the encoder versus using a Leaky ReLU baseline. The model will use the Adam optimizer and MSE loss.
Evaluation Metrics: ['Qualitative comparison of generated output images']
Risk Factors and Limitations

- In order to generalization, the individualization is the reason of shake, the accuracy may be good in some set and may be worse in some set.

Scores:

- Soundness: 1.0
- Support Ratio: 1.0
- Rigor Bucket: low

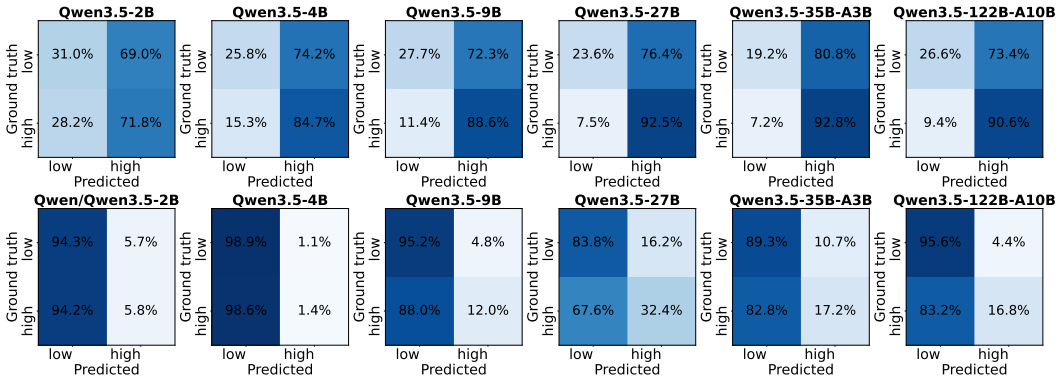

Figure 5: Confusion matrices across six Qwen3.5 model sizes (2B–122B) under standard **(top)** and aggressive **(bottom)** prompting. Under standard prompting, high-soundness recall improves with scale but low-soundness recall degrades. Larger models become more optimistic. Under aggressive prompting, models shift toward over-conservatism with no consistent improvement from scale.

A.5 Example of Extracted Atomic Claims and Verification

Example: Atomic Claims from Extracted Proposal

Source: Identifiability for Gaussian Processes with Holomorphic Kernels

Atomic Claims (excerpt)

- A novel theoretical framework is based on kernels being holomorphic around zero.
- The framework determines identifiability of parameters in a wide class of Gaussian Process kernels.
- The framework applies to squared exponential, periodic, and rational quadratic kernels.
- The framework applies to complex combinations of kernels.
- The framework addresses a gap left by methods that only apply to Matérn-type kernels.
- A Gaussian Process model is fitted to synthetically generated data with added Gaussian noise.
- Parameters are estimated using Maximum Likelihood Estimators.
- Experiments evaluate convergence of estimators as sample size increases.

Summary: 17 atomic claims extracted; all verified as supported (support ratio = 1.0).

B Supplementary Material for Evaluation

B.1 Does Scale Resolve the Optimism Bias?

To isolate the effect of model scale from other factors such as model family or training recipe, we evaluate six models from the same Qwen3.5 family, ranging from 2B to 122B parameters, under both prompting conditions. Results are shown in Fig. 5.

Under standard prompting, scaling improves high-soundness recall monotonically (71.8% at 2B to 92.8% at 35B), but low-soundness recall simultaneously decreases (31.0% at 2B to 19.2% at 35B). Larger models become *more* permissive toward weak proposals, not less. This suggests that scale improves general language understanding and pattern recognition, but does not correct the optimism tendency in scientific judgment.

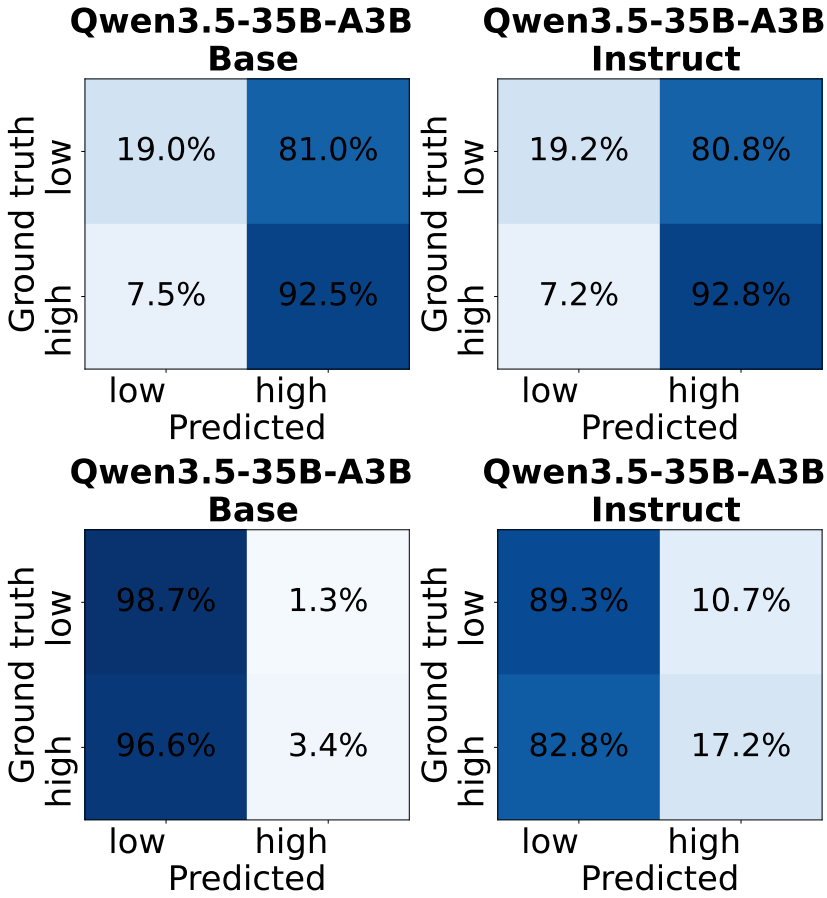

Figure 6: Base vs. instruction-tuned Qwen3.5-35B-A3B under standard **(top)** and aggressive **(bottom)** prompting. Under standard prompting, both variants show similar optimism bias, consistent with an origin earlier in training. Under aggressive prompting, the instruct model is slightly better, but both remain over-conservative.

Under aggressive prompting, scale provides no consistent benefit. All six models trend toward near-always-low behavior, with high-soundness recall ranging from 1.4% (4B) to 32.4% (27B), showing no monotonic trend with size. Notably, the 27B model achieves better balance than both smaller and larger variants, suggesting that the aggressive prompt interacts non-monotonically with model capacity.

Together, these results indicate that the optimism bias observed in Sec. 3.2 is not a small-model artifact in this family. It persists with scale under standard prompting and remains unresolved under aggressive prompting at all tested sizes. Reliable scientific soundness judgment likely requires interventions beyond simply using larger models.

B.2 Is the Optimism Bias Caused by Instruction Tuning?

A natural hypothesis is that the optimism bias stems from instruction tuning or RLHF, which may train models to be agreeable and avoid negative feedback. To test this, we compare Qwen3.5-35B-A3B in its base and instruction-tuned variants under both prompting conditions, shown in Fig. 6.

Under standard prompting, the base and instruct models are nearly indistinguishable: both achieve 19.0%/19.2% low-soundness recall and 92.5%/92.8% high-soundness recall, respectively. These similar numbers suggest instruction tuning is unlikely to be the sole driver of the optimism bias.

Under aggressive prompting, both models shift toward over-conservatism, but the base model shifts more strongly (98.7% low recall, 3.4% high recall) than the instruct variant (89.3% low recall, 17.2% high recall). This suggests instruction tuning may provide modest robustness to prompt pressure, but is insufficient to maintain calibrated judgment.

These findings suggest that resolving the optimism bias may require interventions at or before pre-training (such as targeted training on scientific judgment), rather than relying on post-hoc instruction tuning alone.